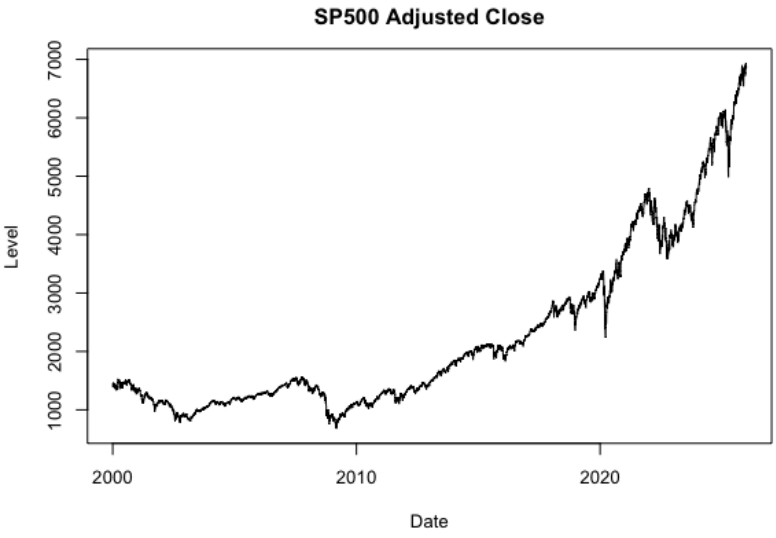
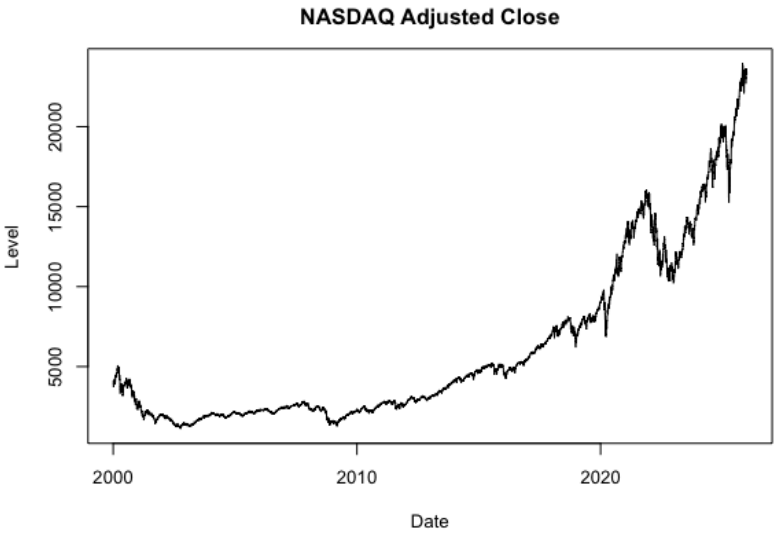
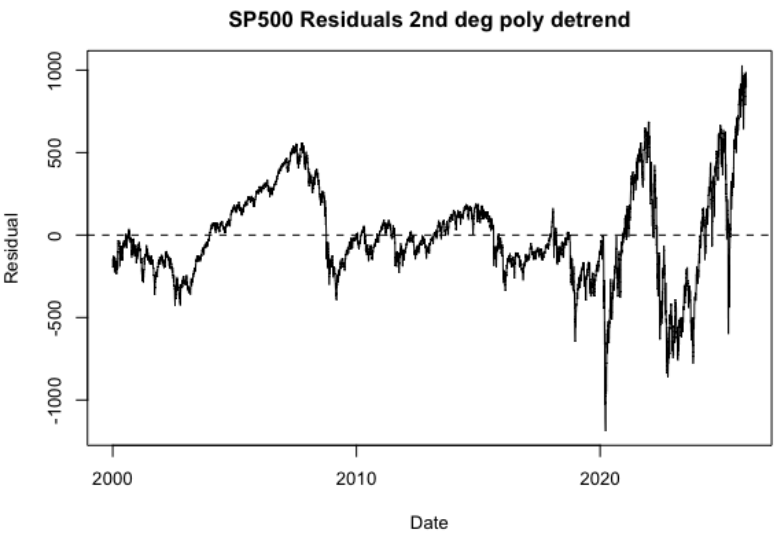
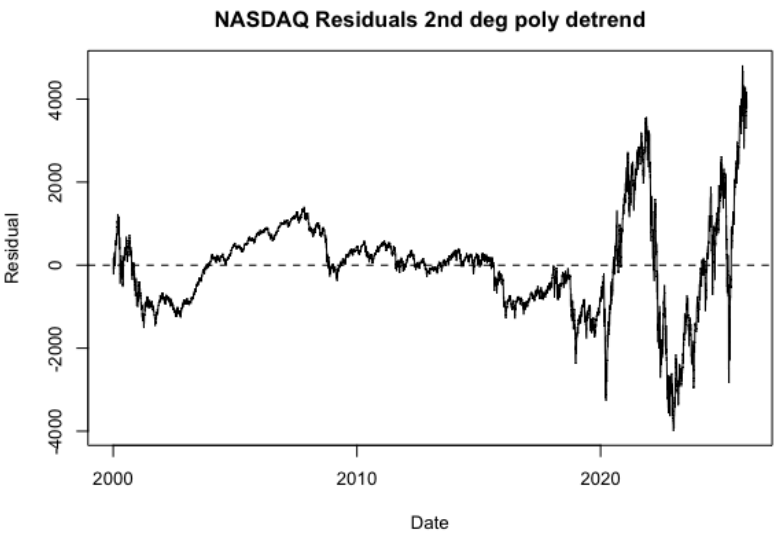
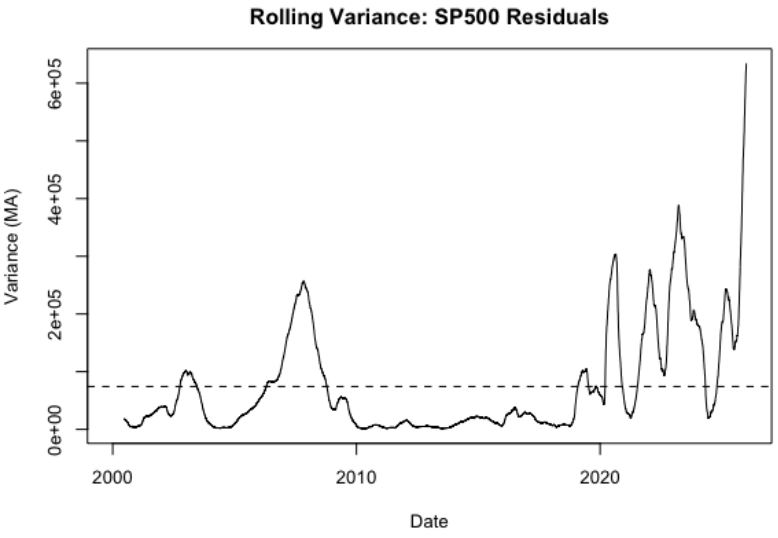
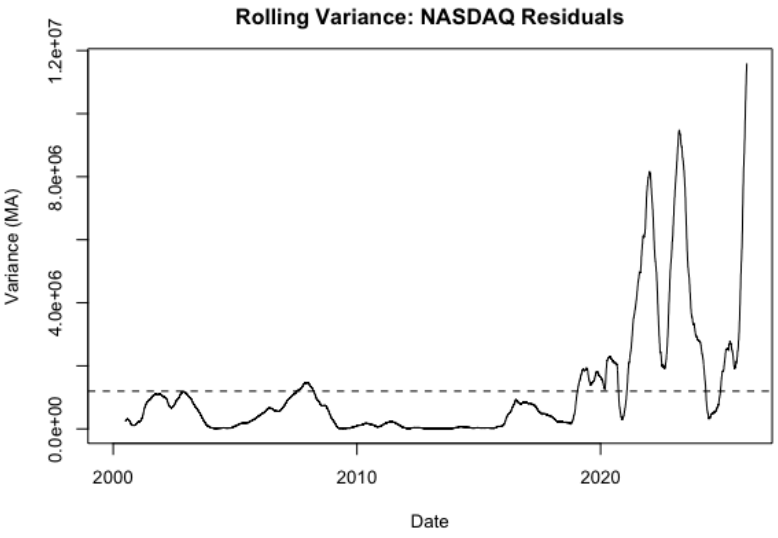

SP500 Adjusted Close
Level
1000
2000
3000
4000
5000
6000
7000
2000
2010
2020
Date

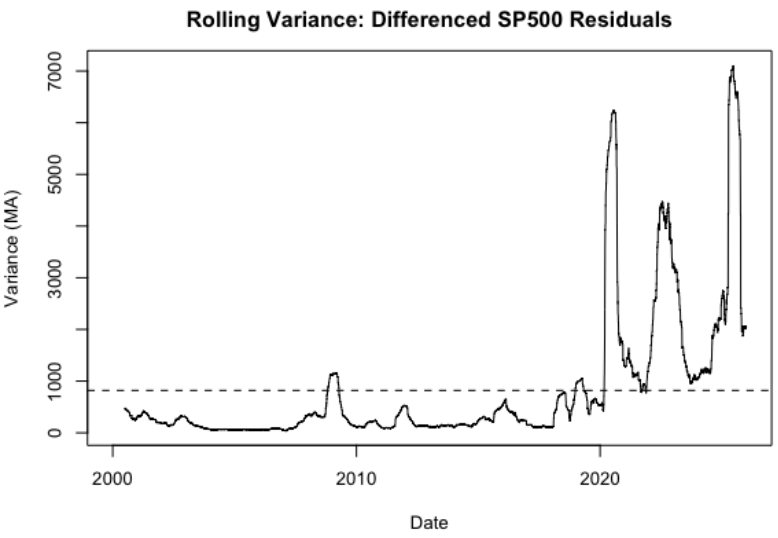

NASDAQ Adjusted Close
Level
5000
10000
15000
20000
2000
2010
2020
Date

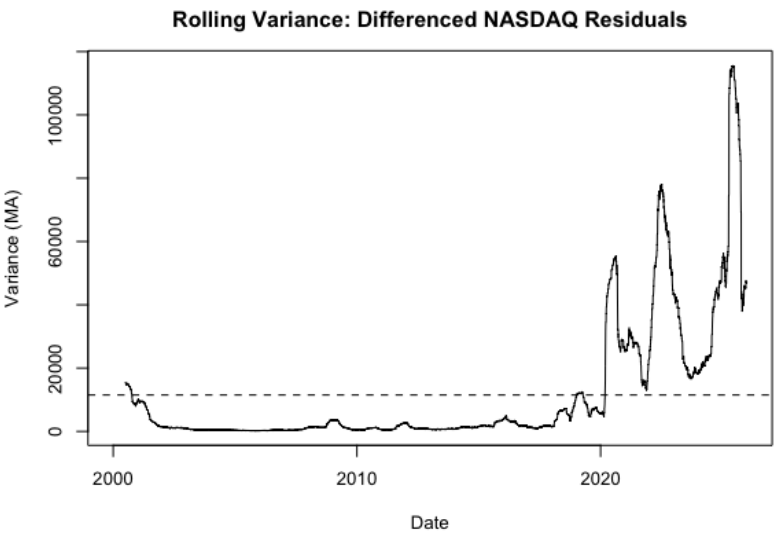

SP500 Residuals 2nd deg poly detrend
Residual
-1000
-500
0
500
1000
2000
2010
2020
Date

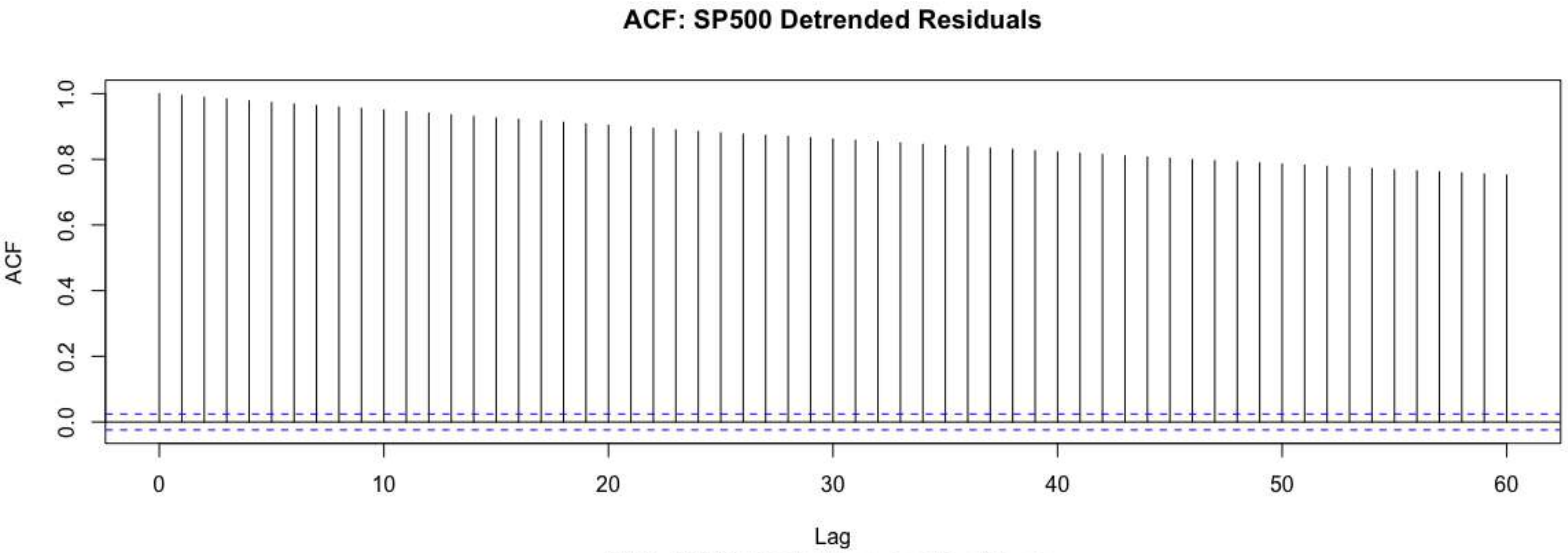

NASDAQ Residuals 2nd deg poly detrend
Residual
-4000
-2000
0
2000
4000
2000
2010
2020
Date

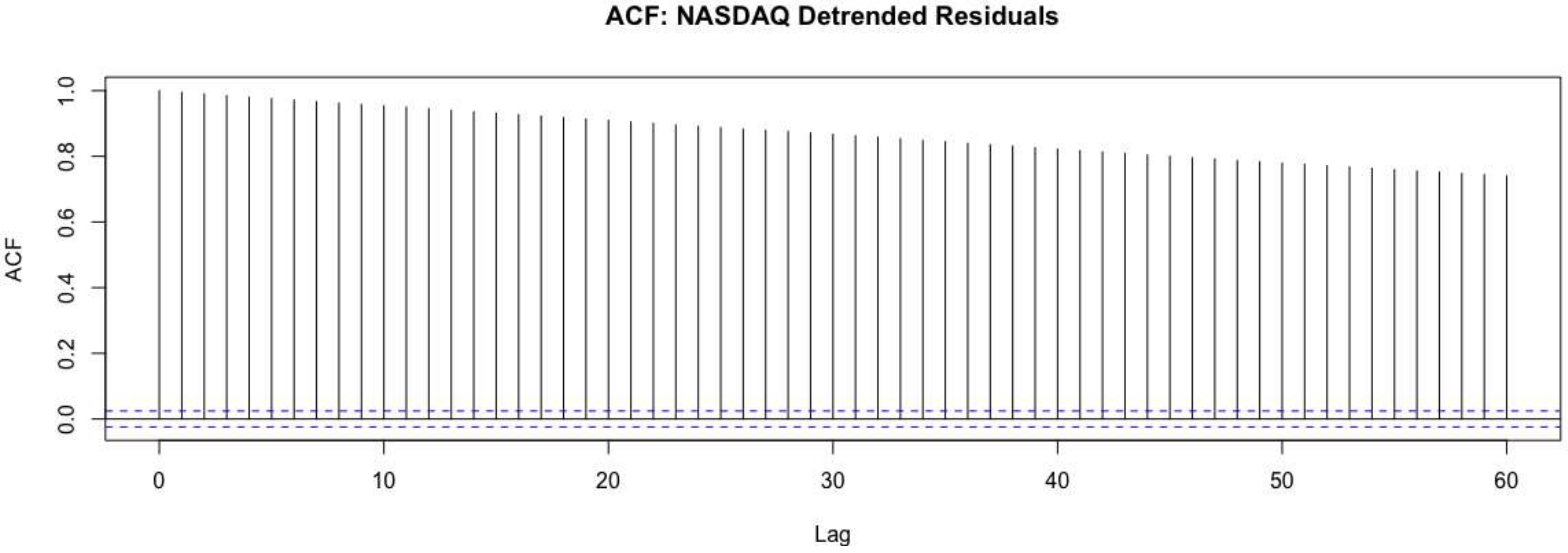

Rolling Variance: SP500 Residuals
Variance (MA)
0e+00
2e+05
4e+05
6e+05
2000
2010
2020
Date

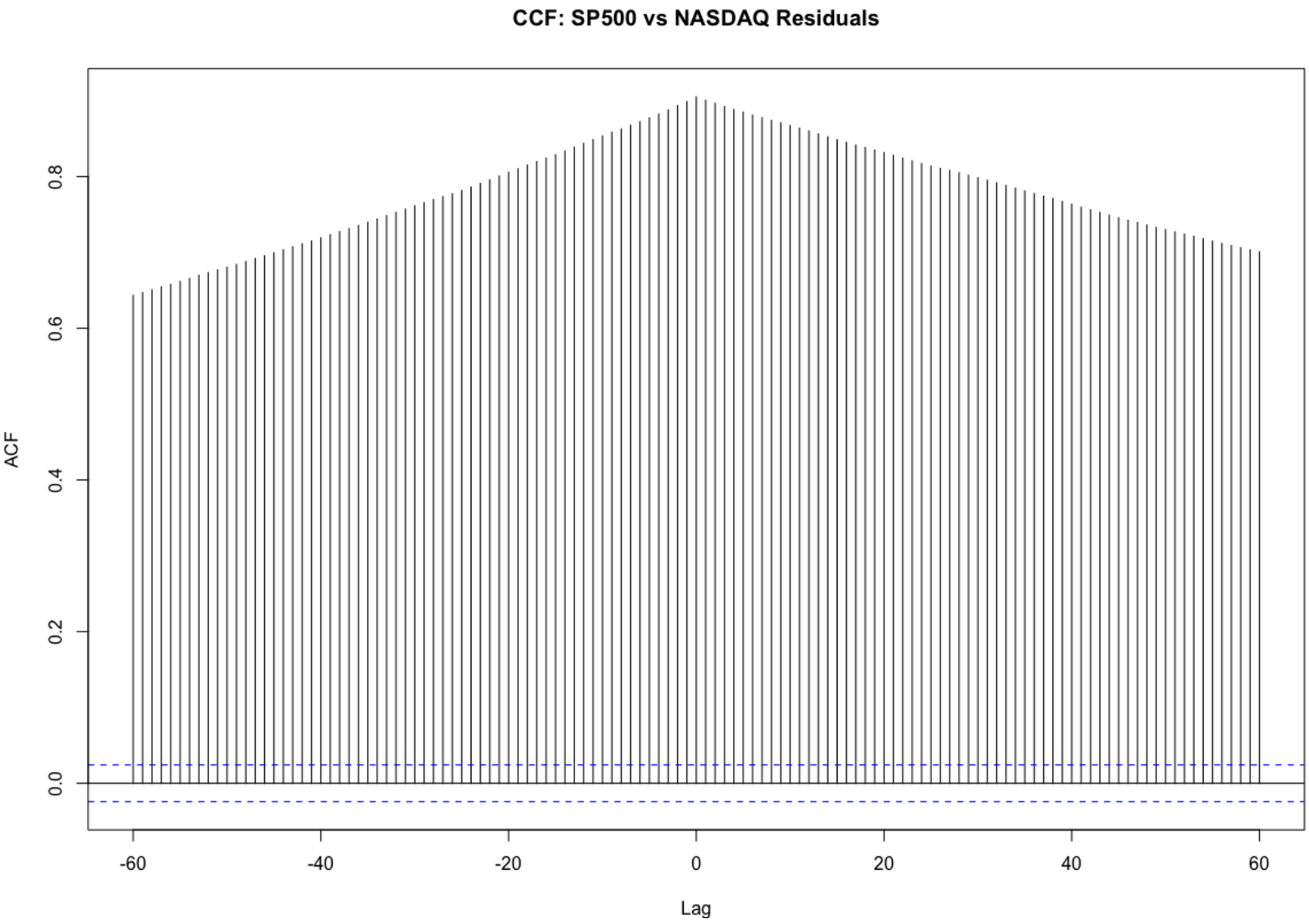

Rolling Variance: NASDAQ Residuals
Variance (MA)
0.0e+00
4.0e+06
8.0e+06
1.2e+07
2000
2010
2020
Date


Rolling Variance: Differenced SP500 Residuals
Variance (MA)
0
1000
3000
5000
7000
2000
2010
2020
Date


Rolling Variance: Differenced NASDAQ Residuals
Variance (MA)
0
20000
60000
100000
2000
2010
2020
Date


ACF: SP500 Detrended Residuals
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
60
Lag


ACF: NASDAQ Detrended Residuals
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
60
Lag


CCF: SP500 vs NASDAQ Residuals
ACF
0.0
0.2
0.4
0.6
0.8
-60
-40
-20
0
20
40
60
Lag